

ATPL EASA STUDY GUIDE | 010 - AIR LAW

Chapter 12: SSR and ACAS (TCAS II)

Syllabus Source: CAE Oxford (pp. 253-262) | Pass Target: >= 90% AviationExam

1. SECONDARY SURVEILLANCE RADAR (SSR) MODES

- Mode A: 4-digit octal code (4,096 combinations) for aircraft identification assigned by ATC.
- Mode C: Transmits automated pressure altitude (referenced to 1013.25 hPa) in 100 ft increments.
- Mode S: Transmits unique 24-bit aircraft address; selective interrogation; enhanced surveillance parameters (Downlink Airborne Parameters - DAP: selected altitude, roll angle, IAS, Mach).

[!] EXAM TRAP (AVIATIONEXAM) - Mandatory Emergency Transponder Codes (Memorize 100%)

- 7500: Unlawful Interference (Hijacking). - 7600: Lost Communications (Radio Failure). - 7700: General Emergency / Distress. - 7000: Standard VFR conspicuity in Europe (1200 in USA). - 2000: Default IFR entry code into uncontrolled airspace or oceanic control without assigned code.

2. AIRBORNE COLLISION AVOIDANCE SYSTEM (ACAS II / TCAS II)

ACAS II interrogates Mode C/S transponders of nearby aircraft to provide warnings based on Time to CPA (tau):

Warning Level	Display Symbol	Time to CPA (Tau)	Cockpit Action & Pilot Response
Traffic Advisory (TA)	Amber solid circle	20 to 48 seconds to CPA	Audio: 'TRAFFIC, TRAFFIC'. Visual scan only. NO EVASIVE MANEUVER PERMITTED.
Resolution Advisory (RA)	Red solid square	15 to 35 seconds to CPA	Audio: 'CLIMB', 'DESCEND', etc. Vertical evasive instruction. Pilot MUST disconnect autopilot and comply immediately.

[!] EXAM TRAP (AVIATIONEXAM) - Pilot Response Time & Absolute Priority of TCAS RA

1. Response Time: Pilot must initiate response within 5 SECONDS for initial RA, and within 2.5 SECONDS for an increased or reversal RA (G-load 0.25 g initial, 0.35 g reversal). 2. RA vs ATC Clearance: TCAS RA OVERRULES ATC CLEARANCE. If an RA contradicts an ATC instruction, the pilot MUST follow the RA without hesitation, and advise ATC as soon as possible ('TCAS RA').

3. TCAS II VERSION 7.1 'LEVEL OFF' MODIFICATION

TCAS II Version 7.1 introduced a critical safety enhancement: the 'Adjust Vertical Speed, Adjust' RA was replaced by 'Level Off, Level Off'. This instructs the crew to reduce vertical climb/descent to zero ft/min, preventing excessive vertical excursions during crossing encounters.

4. MODE S DOWNLINK AIRBORNE PARAMETERS (DAPS)

- Basic DAP: Flight status (airborne/ground), Mode A code, 24-bit aircraft address, aircraft identification (callsign).
- Enhanced DAP: Selected altitude, roll angle, true track angle, ground speed, magnetic heading, indicated airspeed (IAS) / Mach, and vertical rate (barometric or inertial).

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What is the transponder code for an aircraft experiencing unlawful interference (hijacking)?

- [A]: 7700
- [B]: 7600
- [C]: 7500
- [D]: 7000

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. 7500 = hijacking; 7600 = radio failure; 7700 = general emergency.

Q2: In the event of a TCAS Resolution Advisory (RA) contradicting an ATC instruction, what action is required?

- [A]: Follow ATC instruction and ignore TCAS.
- [B]: Ask ATC for clarification before maneuvering.
- [C]: Follow TCAS RA immediately and advise ATC as soon as possible.
- [D]: Turn 90° horizontally to avoid conflict.

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. TCAS RA takes absolute priority over ATC clearances. Comply immediately with RA.

Q3: What is the maximum pilot reaction time allowed to respond to an initial TCAS Resolution Advisory?

- [A]: 2.5 seconds
- [B]: 5.0 seconds
- [C]: 10.0 seconds
- [D]: 15.0 seconds

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Reaction time must be within 5 seconds for initial RA, and within 2.5 seconds for an increased or reversal RA.

Q4: Which transponder code should be set by an IFR aircraft entering an airspace region without an assigned code?

- [A]: 7000
- [B]: 2000
- [C]: 1200
- [D]: 7700

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Mode A Code 2000 is the standard IFR code when no code has been assigned by ATC.

Q5: Under TCAS II Version 7.1, what new resolution advisory replaced 'Adjust Vertical Speed, Adjust'?

- [A]: 'Maintain Vertical Speed'
- [B]: 'Level Off, Level Off'
- [C]: 'Descend Immediately'
- [D]: 'Climb and Turn Left'

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. TCAS II version 7.1 replaced 'Adjust Vertical Speed' with 'Level Off, Level Off' to eliminate pilot over-corrections.

Q6: What is the required pilot response time for an increased or reversal TCAS Resolution Advisory (RA)?

- [A]: 5.0 seconds
- [B]: 2.5 seconds
- [C]: 1.0 second
- [D]: 10.0 seconds

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Pilot reaction time for a reversal or increased rate RA is 2.5 seconds (with 0.35 g target acceleration).